Audio course on steroids during pregnancy. Episode one. The steroid paradox: mother, fetus, or adrenal axis.

Companion reading handout for simultaneous listening.
This course is based on the steroids during pregnancy study guide, Gabbe chapters on placental anatomy, developmental origins, preterm labor, PPROM, preeclampsia, respiratory disease, adrenal disease, hematologic disease, collagen vascular disease, gastrointestinal disease, and skin disease, the Israeli position papers and local protocols on preterm labor, PPROM, late preterm steroids, and diabetes, ACOG guidance on antenatal corticosteroids, BSR guidance on corticosteroids in rheumatologic disease, the global IBD pregnancy consensus, UKTIS safety guidance, and the ALPS trial.

The first job of this course is to make one thing clear. Steroids in pregnancy are not one intervention.

That is the source of the confusion.

One patient receives betamethasone because the fetus is at risk of preterm birth and we want the drug to cross the placenta. Another patient receives prednisone because she has an asthma exacerbation or lupus flare and we do not particularly want fetal steroid exposure. A third patient receives hydrocortisone because she has adrenal insufficiency and labor is a physiologic stress that could cause adrenal crisis.

Those are not variations of the same decision. They are different clinical acts.

So the wrong opening question is, are steroids safe in pregnancy?

That question is too blunt to be useful. A better question is, who is the target?

Is the target the fetus? Is the target the mother? Or is the target the maternal adrenal axis?

Once you name the target, the rest of the decision becomes much more coherent. Which steroid should be used? How much should be given? For how long? At what gestational age? What are we trying to prevent? What are we willing to expose the fetus or neonate to? And what monitoring must be built into the order?

Pause here, because this is the central model for the whole course.

The safest steroid is the one whose placental transfer, dose, timing, duration, and target match the clinical problem.

Let us build the three categories.

The first category is fetal-benefit steroids.

Here, the fetus is the target. We choose betamethasone or dexamethasone because they cross the placenta. That is not a side effect. That is the point. We want fetal tissues to see the steroid because the goal is accelerated maturation, especially of the lung, before likely preterm birth.

This is the usual antenatal corticosteroid story. Betamethasone twelve milligrams intramuscularly, repeated once after twenty four hours. Or dexamethasone six milligrams intramuscularly every twelve hours for four doses. These regimens are not simply drug trivia. They represent a very specific fetal therapy given in a very specific time window.

The second category is maternal disease-control steroids.

Here, the mother is the target. The disease may be asthma, lupus, inflammatory bowel disease, pemphigoid gestationis, ITP, severe eczema, vasculitis, or another inflammatory process. In this category, fetal exposure is usually not the therapeutic goal. The goal is to control maternal disease while limiting unnecessary fetal exposure.

This is why prednisone, prednisolone, hydrocortisone, and methylprednisolone often appear in maternal disease management. The placenta metabolizes much of these drugs, so fetal exposure is lower than with betamethasone or dexamethasone.

Do not misunderstand that sentence. Lower fetal exposure does not mean zero fetal exposure. It means the pharmacology matches the clinical target better.

The third category is replacement and stress-dose steroids.

Here, the goal is not fetal lung maturation and not anti-inflammatory rescue. The goal is endocrine survival. A patient with adrenal insufficiency, or a patient with clinically important adrenal suppression from prolonged glucocorticoid use, may need hydrocortisone coverage during labor, cesarean delivery, hemorrhage, infection, or surgery.

That is not optional medication exposure in the usual sense. It is replacement of a stress response the patient may not be able to produce.

Now notice what happens if we collapse these categories.

If a patient is taking chronic prednisolone for lupus and then develops real preterm birth risk, a resident may ask, does she already have steroids on board for the fetal lungs?

Usually, no. The reason prednisolone is useful for maternal disease is that much of it is inactivated before reaching the fetus. That same feature means it should not be treated as a reliable fetal lung maturation course.

Now flip the error.

If a patient has a maternal autoimmune flare, a resident may think, dexamethasone is powerful and crosses the placenta, so perhaps it is stronger. But for ordinary maternal disease, crossing the placenta is not an advantage. It may be a disadvantage. Unless the fetus is the intended target, a lower fetal-transfer steroid is usually the cleaner pharmacologic choice.

That is the paradox. Sometimes placental crossing is the therapy. Sometimes placental crossing is the exposure we try to reduce.

This course will keep returning to that distinction.

In fetal lung maturation, the question is timing. Is preterm birth likely soon enough that the fetus benefits? Are we in the gestational window where evidence supports treatment? Are we about to deliver anyway because the mother or fetus is unstable?

In maternal disease, the question is disease control. Is the disease itself dangerous to the pregnancy? Can local, inhaled, or lower fetal-transfer therapy work? Is a steroid-sparing pregnancy-compatible medication needed so the patient does not live on repeated rescue courses?

In adrenal disease, the question is physiologic coverage. Is this patient able to mount a cortisol response to labor or illness? If not, the danger is adrenal crisis.

This framework also changes counseling.

A patient with threatened preterm birth at thirty weeks should hear, this steroid is meant to reach the baby. We are using it because birth may happen soon, and the benefit is strongest when delivery occurs in the next week. We will watch your glucose, and the neonatology team will watch the baby's glucose after birth.

A patient with a lupus flare at ten weeks should hear something different. Here the steroid is for you. We usually choose a prednisolone-type steroid because the placenta breaks down much of it before it reaches the fetus. The goal is to control the flare with the lowest effective dose, because uncontrolled lupus can be dangerous for both you and the pregnancy.

A patient with adrenal insufficiency in labor should hear something different again. This is stress coverage. Labor is a physiologic stress, and your body may not be able to produce enough cortisol. Hydrocortisone prevents a crisis.

Three patients. Three steroid decisions. Three different targets.

The resident reflex is this: before choosing a steroid in pregnancy, name the target.

- If the target is the fetus, choose a steroid that reaches the fetus and give it in the right window.
- If the target is the mother, choose the lowest effective maternal therapy, usually with lower fetal transfer when possible, and do not undertreat serious disease.
- If the target is the adrenal axis, think replacement and stress physiology, not immunosuppression. That is the language of the course. Steroids are not safe or unsafe as a class. Steroids are matched or mismatched to a target. The rest of the course is about learning to make that match with precision.